In [1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

%matplotlib inline
```

In [2]:
```
data = pd.read_csv('Iris_Dataset.csv')
data
```

Out[2]:
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Id</th> <th>SepalLengthCm</th> <th>SepalWidthCm</th> <th>PetalLengthCm</th> <th>PetalWidthCm</th> <th>variety</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>...</th> <th>145</th> <th>146</th> <th>147</th> <th>148</th> <th>149</th> </tbody> <tbody></tbody>

1	5.1	3.5	1.4	0.2	Iris-setosa
2	4.9	3.0	1.4	0.2	Iris-setosa
3	4.7	3.2	1.3	0.2	Iris-setosa
4	4.6	3.1	1.5	0.2	Iris-setosa
5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
146	6.7	3.0	5.2	2.3	Iris-virginica
147	6.3	2.5	5.0	1.9	Iris-virginica
148	6.5	3.0	5.2	2.0	Iris-virginica
149	6.2	3.4	5.4	2.3	Iris-virginica
150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

In [3]:
```
data.head()
```

Out[3]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Id</th> <th>SepalLengthCm</th> <th>SepalWidthCm</th> <th>PetalLengthCm</th> <th>PetalWidthCm</th> <th>variety</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

1	5.1	3.5	1.4	0.2	Iris-setosa
2	4.9	3.0	1.4	0.2	Iris-setosa
3	4.7	3.2	1.3	0.2	Iris-setosa
4	4.6	3.1	1.5	0.2	Iris-setosa
5	5.0	3.6	1.4	0.2	Iris-setosa

In [4]:
```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   variety        150 non-null    str
dtypes: float64(4), int64(1), str(1)
memory usage: 7.2 KB
```

In [5]:
```
data.describe()
```

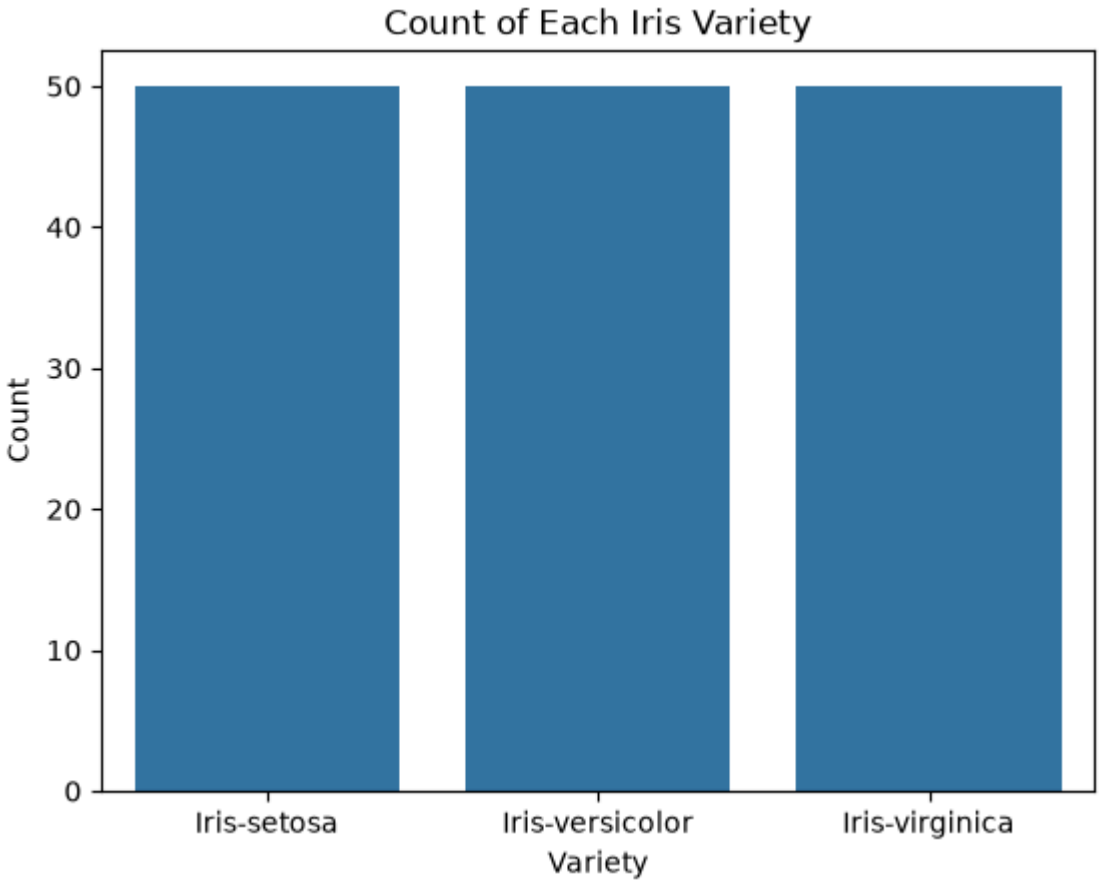
Out[5]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Id</th> <th>SepalLengthCm</th> <th>SepalWidthCm</th> <th>PetalLengthCm</th> <th>PetalWidthCm</th> </thead> <tbody> <th>count</th> <th>mean</th> <th>std</th> <th>min</th> <th>25%</th> <th>50%</th> <th>75%</th> <th>max</th> </tbody> <tbody></tbody>

150.000000	150.000000	150.000000	150.000000	150.000000
75.500000	5.843333	3.057333	3.758000	1.199333
43.445368	0.828066	0.435866	1.765298	0.762238
1.000000	4.300000	2.000000	1.000000	0.100000
38.250000	5.100000	2.800000	1.600000	0.300000
75.500000	5.800000	3.000000	4.350000	1.300000
112.750000	6.400000	3.300000	5.100000	1.800000
150.000000	7.900000	4.400000	6.900000	2.500000

In [6]:
```
data.value_counts('variety')
```

Out[6]:
```
variety
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```

In [7]:
```
sns.countplot(x='variety', data=data)
plt.title('Count of Each Iris Variety')
plt.xlabel('Variety')
plt.ylabel('Count')
plt.show()
```



In [8]:
```
dummies = pd.get_dummies(data['variety'])
dummies
```

Out[8]:

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Iris-setosa</th> <th>Iris-versicolor</th> <th>Iris-virginica</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>...</th> <th>145</th> <th>146</th> <th>147</th> <th>148</th> <th>149</th> </tbody> <tbody></tbody>

0	True	False	False
1	True	False	False
2	True	False	False
3	True	False	False
4	True	False	False
...	...	...	...
145	False	False	True
146	False	False	True
147	False	False	True
148	False	False	True
149	False	False	True

150 rows × 3 columns

In [9]:

```
FinalDataset = pd.concat(
    [dummies, data.iloc[:, [0, 1, 2, 3]]],
    axis=1
)

FinalDataset.head()
```

Out[9]:

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Iris-setosa</th> <th>Iris-versicolor</th> <th>Iris-virginica</th> <th>Id</th> <th>SepalLengthCm</th> <th>SepalWidthCm</th> <th>PetalLengthCm</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

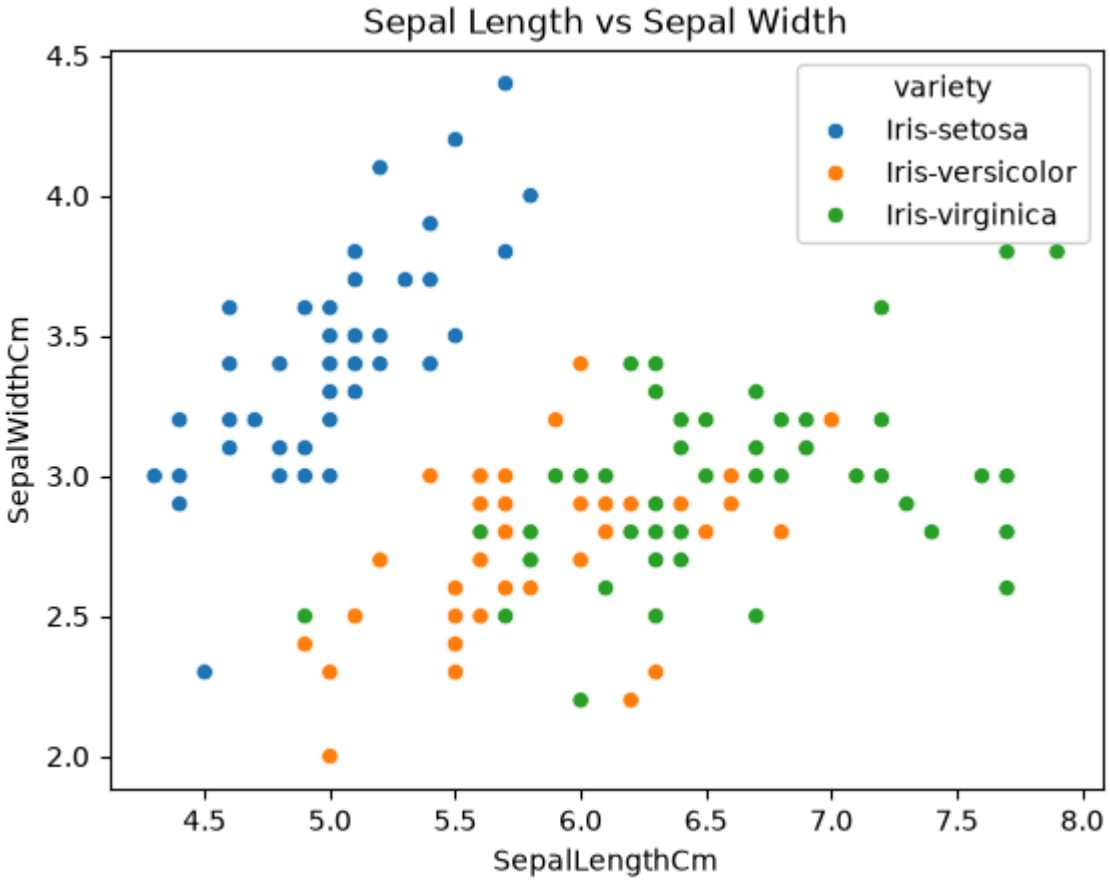
0	True	False	False	1	5.1	3.5	1.4
1	True	False	False	2	4.9	3.0	1.4
2	True	False	False	3	4.7	3.2	1.3
3	True	False	False	4	4.6	3.1	1.5
4	True	False	False	5	5.0	3.6	1.4

In [10]:

```
sns.scatterplot(
    x='SepalLengthCm',
    y='SepalWidthCm',
```

```
    hue='variety',
    data=data
)

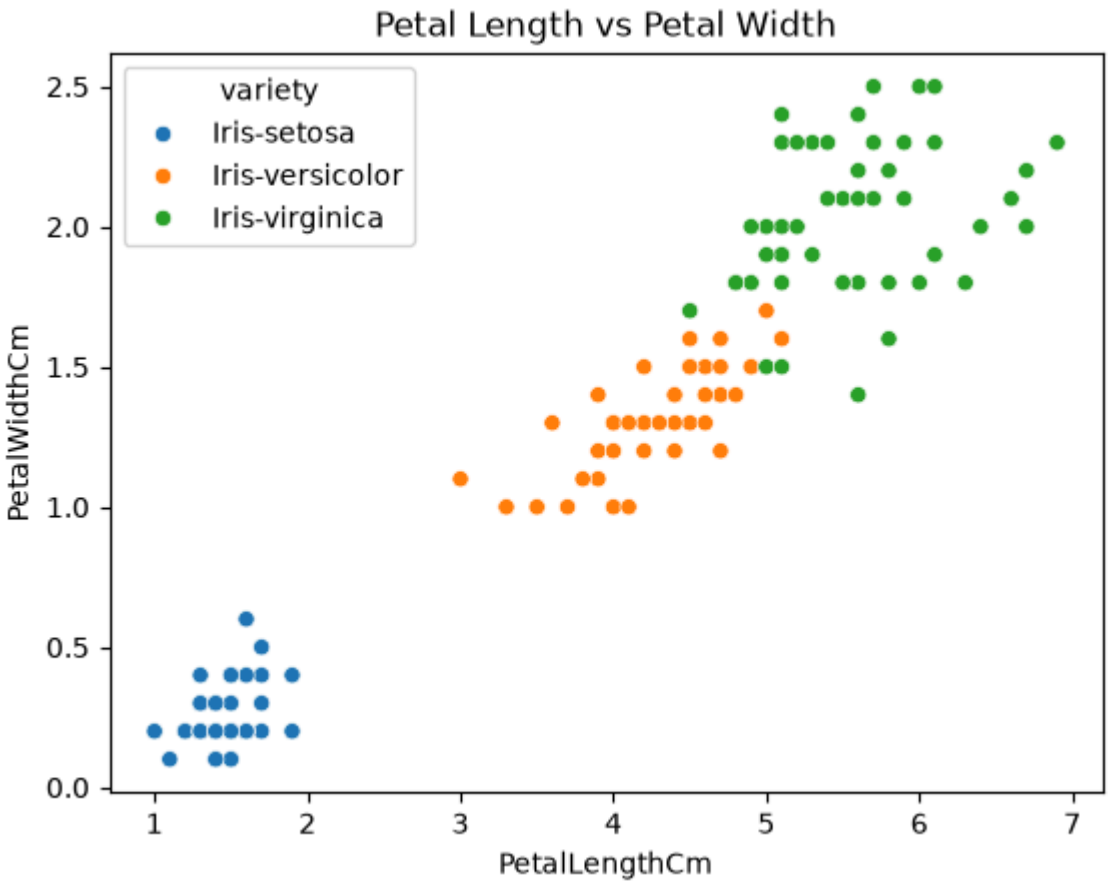
plt.title('Sepal Length vs Sepal Width')
plt.show()
```



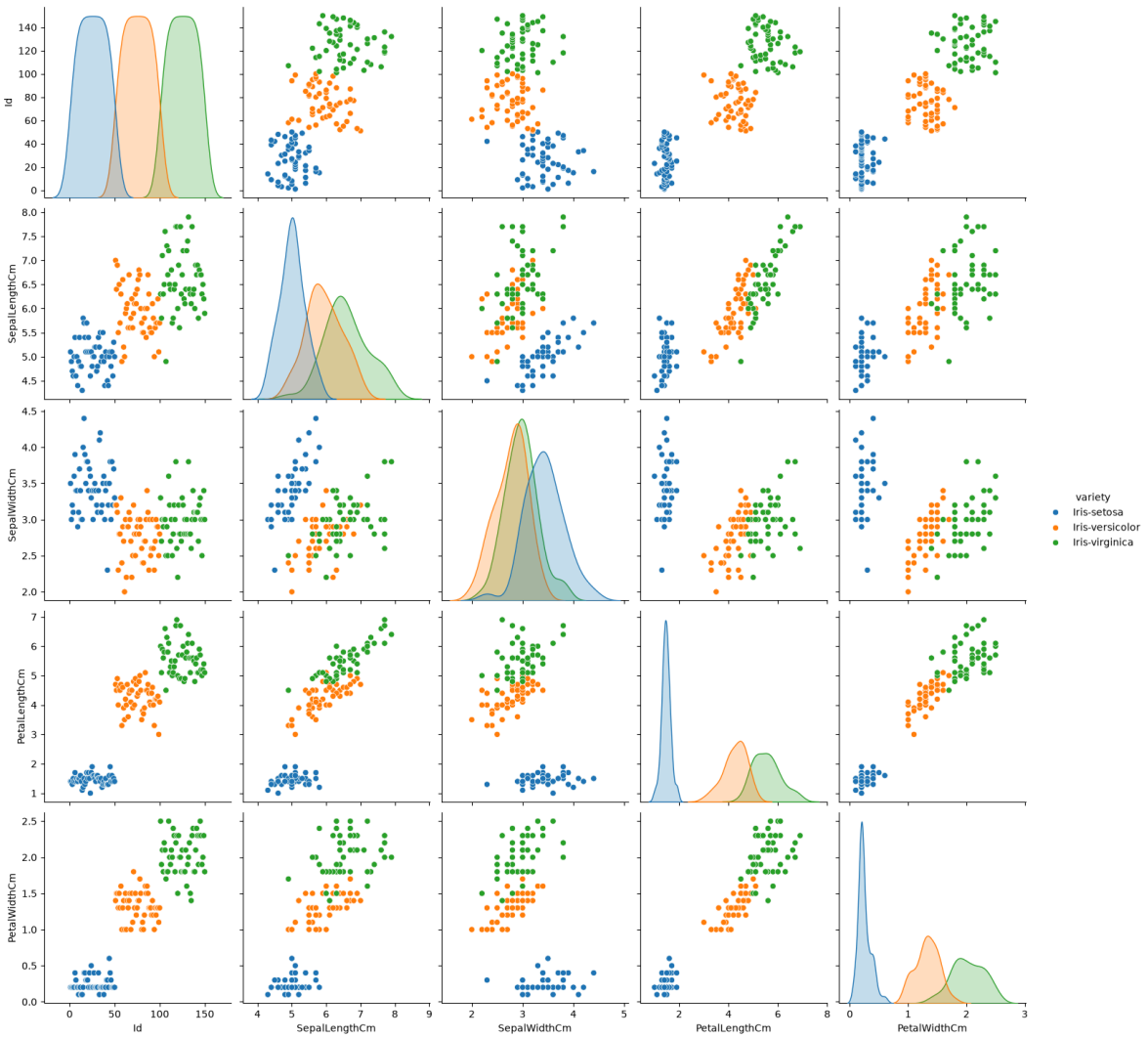
In [11]:

```
sns.scatterplot(
    x='PetalLengthCm',
    y='PetalWidthCm',
    hue='variety',
    data=data
)

plt.title('Petal Length vs Petal Width')
plt.show()
```

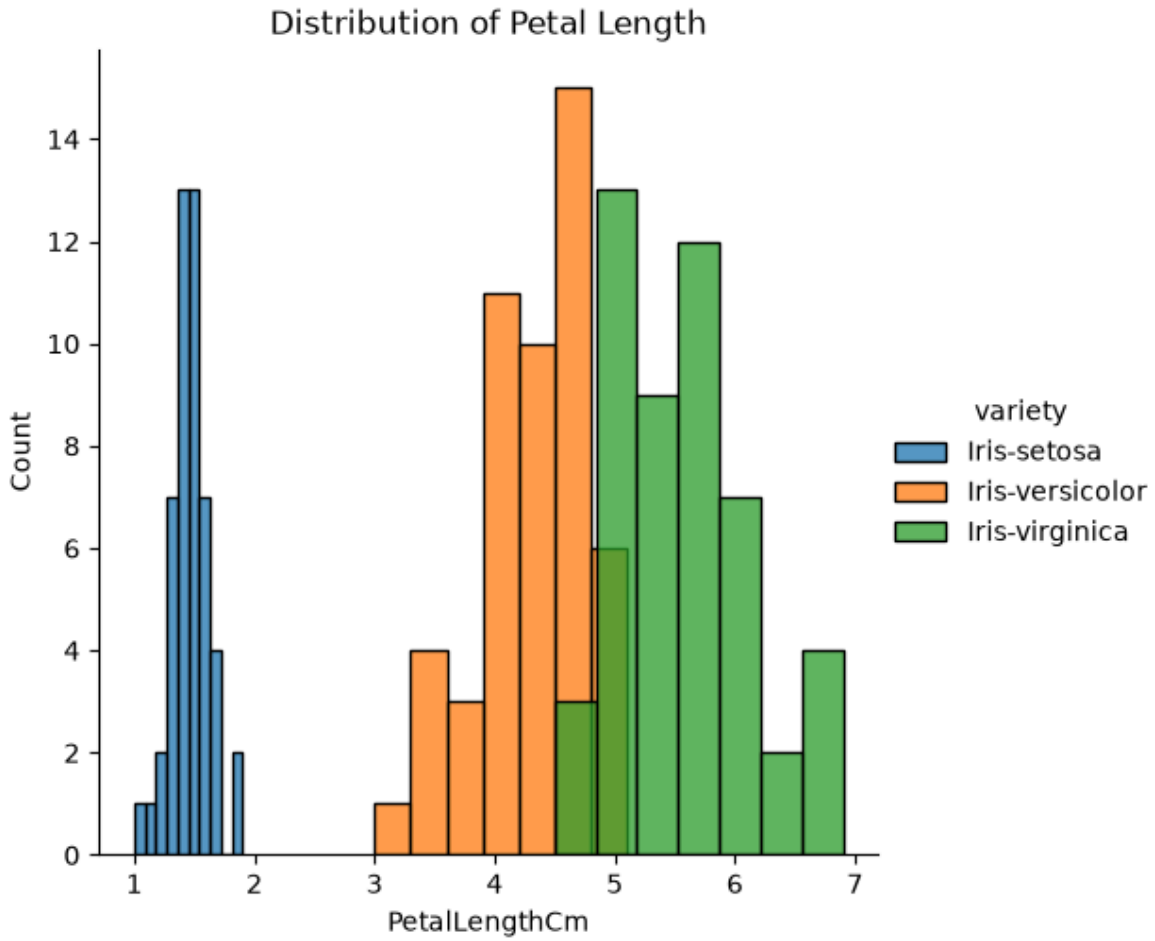


In [12]:
```
sns.pairplot(data, hue='variety', height=3)
plt.show()
```



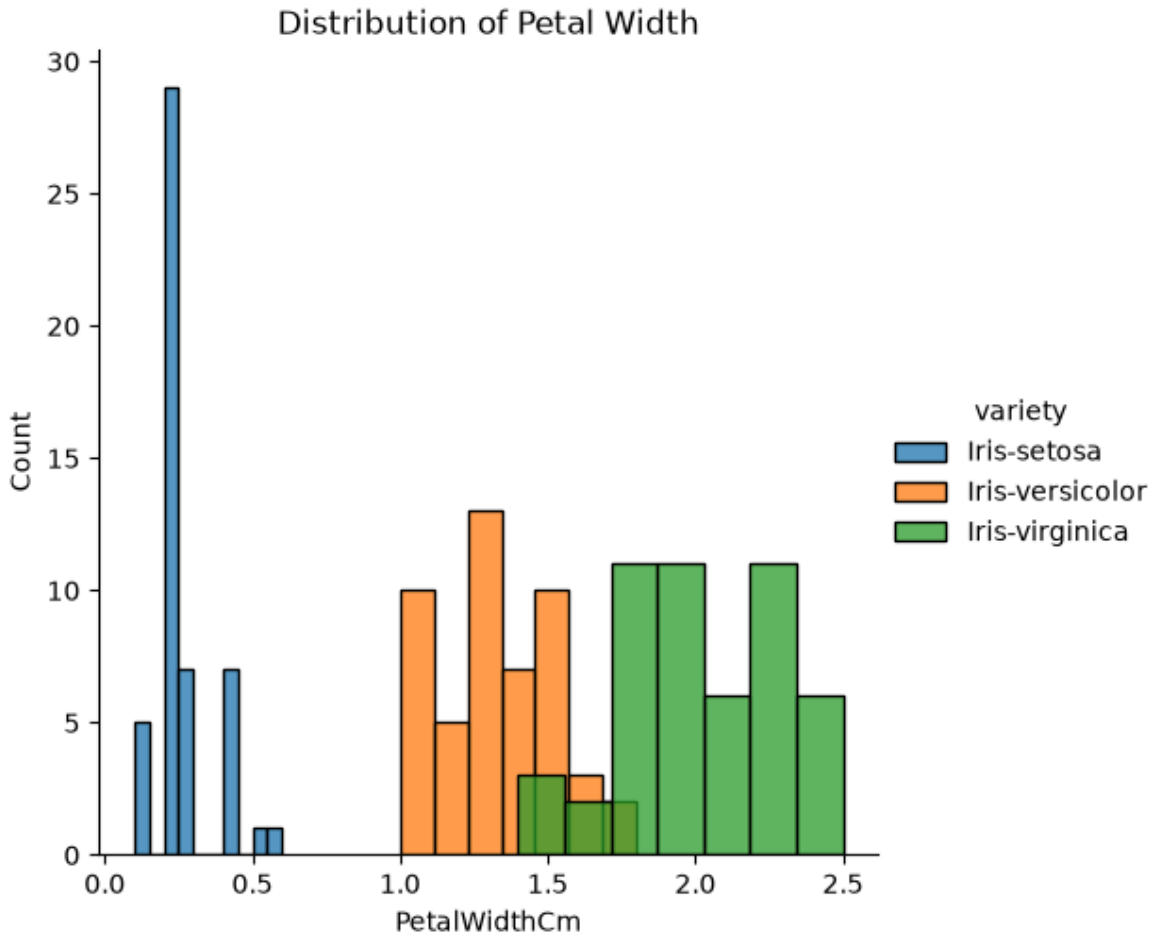
In [13]:
```
sns.FacetGrid(
    data,
    hue='variety',
    height=5
).map(
    sns.histplot,
    'PetalLengthCm'
).add_legend()

plt.title('Distribution of Petal Length')
plt.show()
```



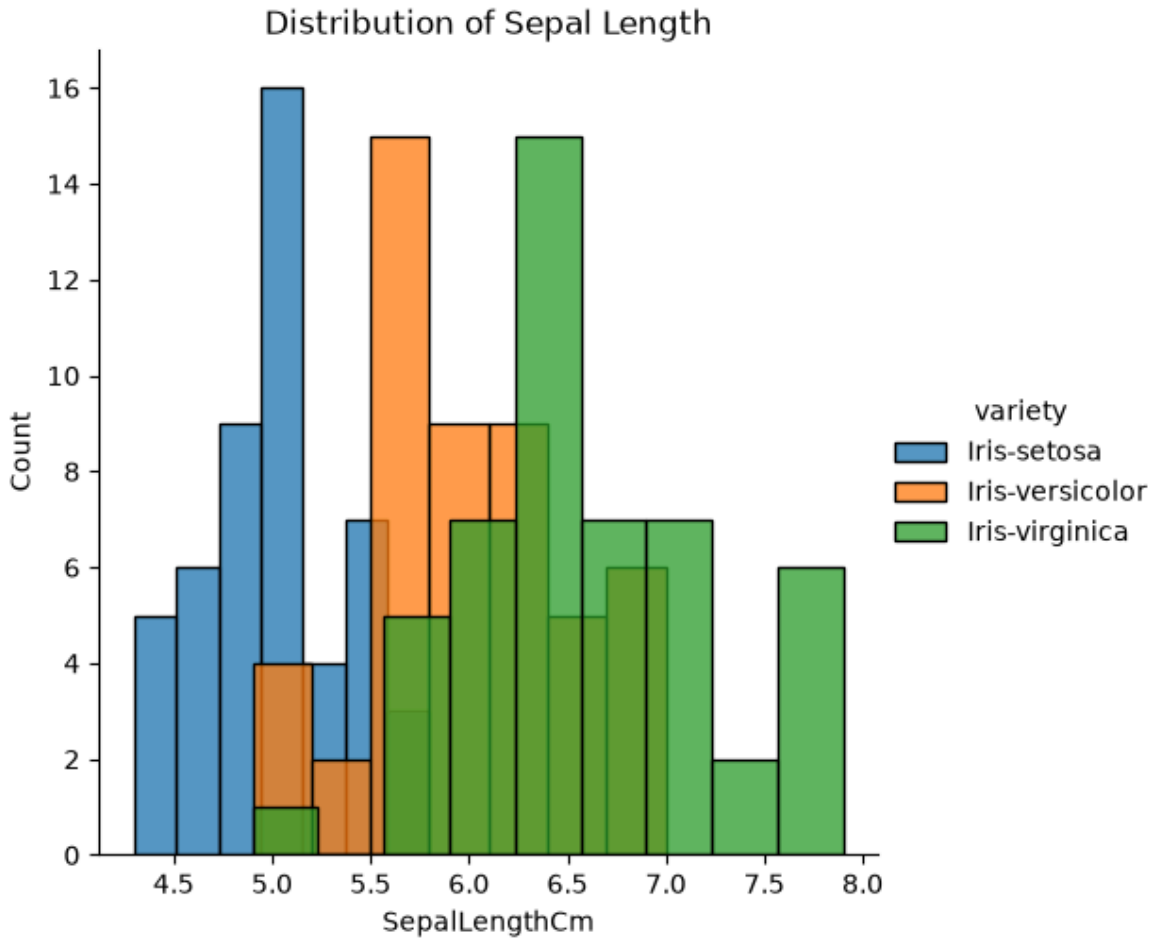
In [14]:
```
sns.FacetGrid(
    data,
    hue='variety',
    height=5
).map(
    sns.histplot,
    'PetalWidthCm'
).add_legend()

plt.title('Distribution of Petal Width')
plt.show()
```



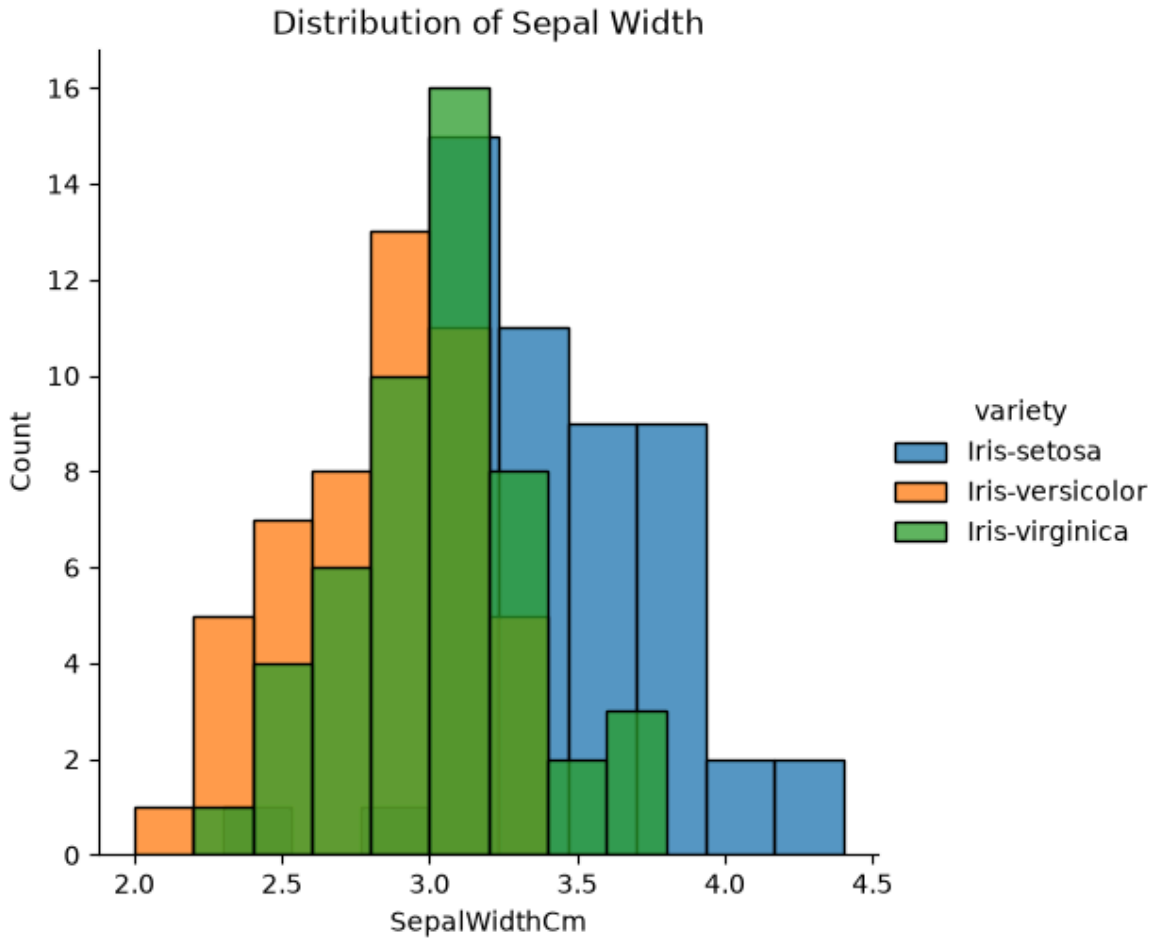
In [15]:
```
sns.FacetGrid(
    data,
    hue='variety',
    height=5
).map(
    sns.histplot,
    'SepalLengthCm'
).add_legend()

plt.title('Distribution of Sepal Length')
plt.show()
```



In [16]:
```
sns.FacetGrid(
    data,
    hue='variety',
    height=5
).map(
    sns.histplot,
    'SepalWidthCm'
).add_legend()

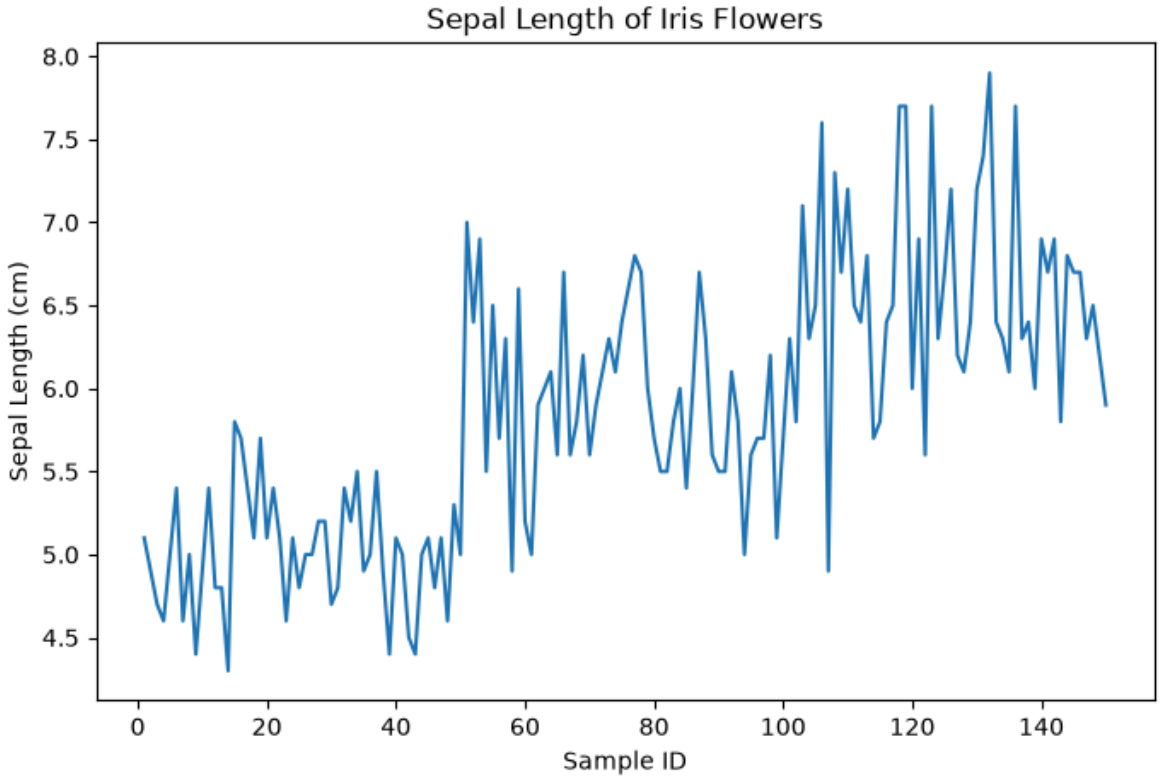
plt.title('Distribution of Sepal Width')
plt.show()
```



In [17]:
```
plt.figure(figsize=(8, 5))

plt.plot(
    data['Id'],
    data['SepalLengthCm']
)

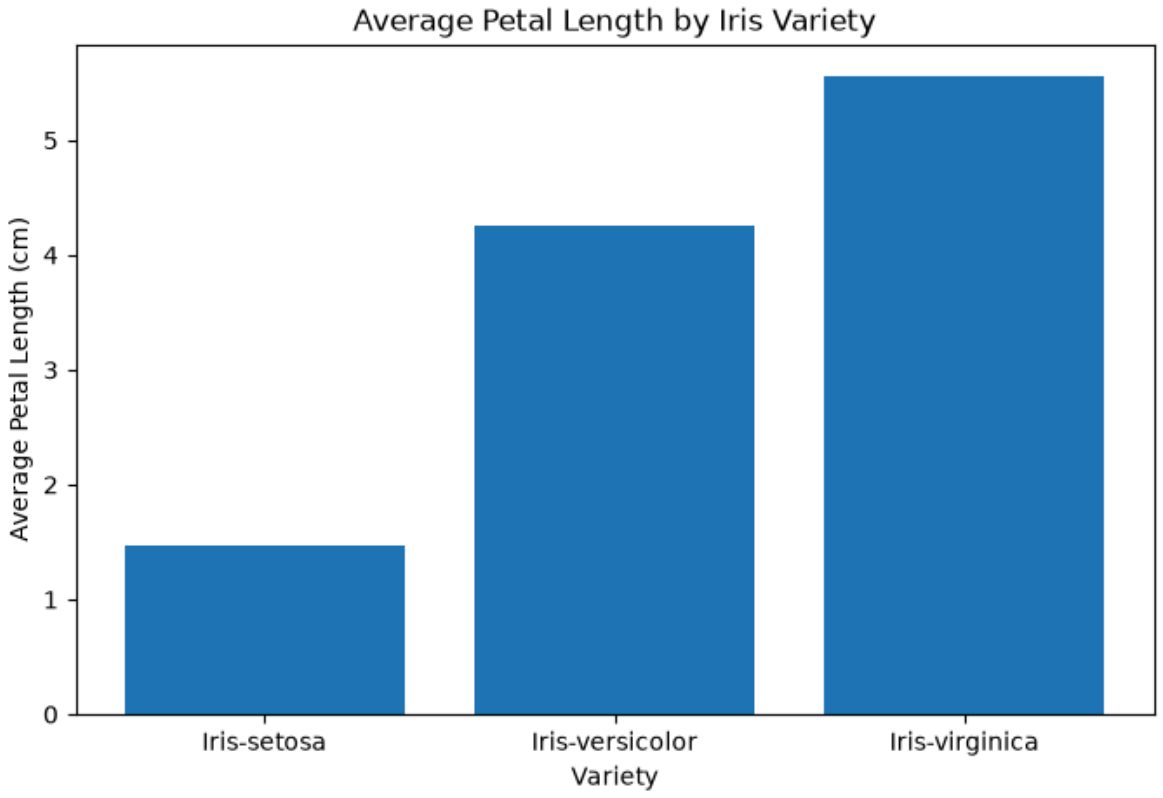
plt.title('Sepal Length of Iris Flowers')
plt.xlabel('Sample ID')
plt.ylabel('Sepal Length (cm)')
plt.show()
```



In [18]:
```
avg_petal = data.groupby('variety')['PetalLengthCm'].mean()

plt.figure(figsize=(8, 5))
plt.bar(avg_petal.index, avg_petal.values)

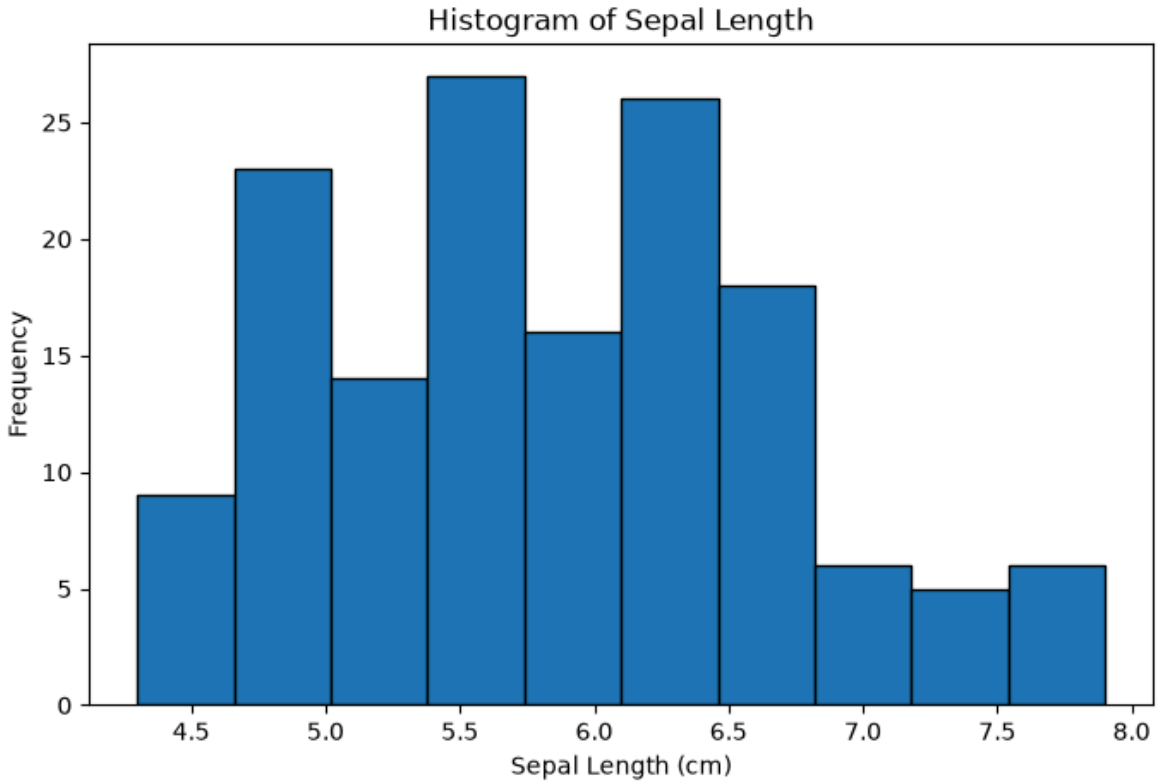
plt.title('Average Petal Length by Iris Variety')
plt.xlabel('Variety')
plt.ylabel('Average Petal Length (cm)')
plt.show()
```



In [19]:
```
plt.figure(figsize=(8, 5))

plt.hist(data['SepalLengthCm'], bins=10, edgecolor='black')

plt.title('Histogram of Sepal Length')
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Frequency')
plt.show()
```

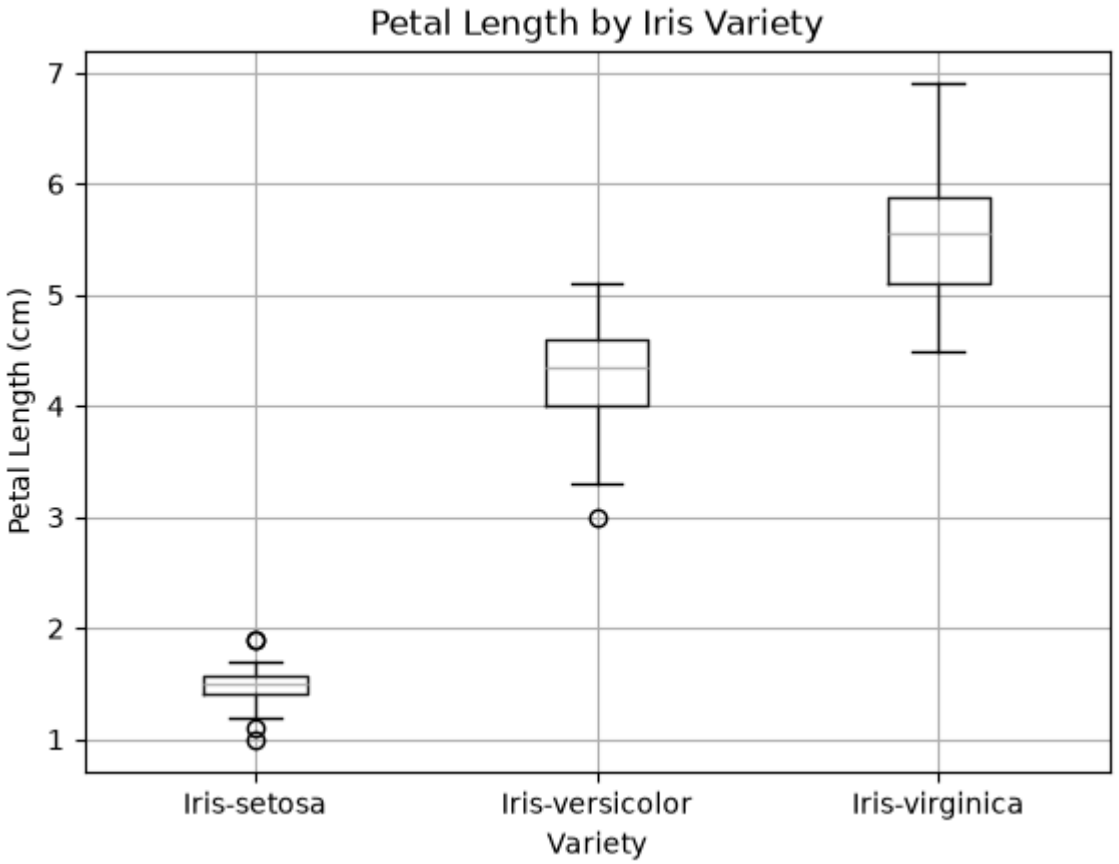


In [20]:
```
plt.figure(figsize=(8, 5))

data.boxplot(column='PetalLengthCm', by='variety')

plt.title('Petal Length by Iris Variety')
plt.suptitle('')
plt.xlabel('Variety')
plt.ylabel('Petal Length (cm)')
plt.show()
```

```
<Figure size 800x500 with 0 Axes>
```


Petal Length by Iris Variety
Petal Length (cm)
7
6
5
4
3
2
1
Iris-setosa
Iris-versicolor
Iris-virginica
Variety

In [ ]: